

CE 310 — Week 4 Lab: When Rare Becomes Routine

Dataset: NOAA Daily Precipitation — Tucson International Airport, 2013–2022 Topic: Poisson Distribution · Event Rate (λ) · PMF · CDF · Return Periods

Before You Begin

Files needed: - Week4_Tucson_Precipitation_Daily_2013_2022.csv — 3,652 daily rows of precipitation data - Week4_Answer_Sheet.xlsx — download from D2L and enter all answers here

Dataset note: This is real station data from NOAA's Climate Data Online for Tucson International Airport (Station USW00023160). Each row is one calendar day, 2013–2022. The column PRCP_in records daily total precipitation in inches; IsRainDay = 1 if any precipitation was recorded, 0 otherwise.

Answer Sheet instructions: Type every numerical answer in Col E of the row matching the question label — A1 in row A1, B3 in row B3. For ★ questions, also enter the intermediate value in Col D. For Col C, type the Excel formula or method you used.

Type your formula as text. In Col C, put an apostrophe before the = so Excel stores the formula instead of running it — for example, '=AVERAGE(B2:B8761)'. Without the apostrophe Excel evaluates it and your method is lost. Screenshots are not accepted; the typed formula is what earns the approach credit.

Background

Today you'll fit λ from real precipitation data and use it to evaluate exactly how rare a “rare” event really is.

Excel Quick Reference

Goal	Formula
Count non-empty cells to verify row count	=COUNTA(A:A)-1 — subtract 1 to exclude the header
Count events in a category	=COUNTIFS(E2:E3653, "Summer", G2:G3653, 1)
Poisson PMF — $P(X = k)$	=POISSON.DIST(k, lambda, FALSE)
Poisson CDF — $P(X \leq k)$	=POISSON.DIST(k, lambda, TRUE)

Goal	Formula
$P(X \geq k)$ — at least k events	$=1 - \text{POISSON.DIST}(k-1, \text{lambda}, \text{TRUE})$
$P(X > k)$ — more than k events	$=1 - \text{POISSON.DIST}(k, \text{lambda}, \text{TRUE})$
Return period formula	$=1 - \text{EXP}(-N/T)$ where N = years, T = return period
Build PMF table	Fill down: $=\text{POISSON.DIST}(A2, \$B\$2, \text{FALSE})$ where $A2$ = k value

Setup (~10 min)

1. Import Week4_Tucson_Precipitation_Daily_2013_2022.csv into a new Excel workbook:
 - Data → Get Data → From Text/CSV → Load.
2. Confirm 3,652 rows of data: $=\text{COUNTA}(A:A)-1$
3. Confirm the 7 column names in row 1:
 - Date, Year, Month, Day, Season, PRCP_in, IsRainDay
 - Season = “Winter” (DJF), “Spring” (MAM), “Summer” (JJA), “Fall” (SON)
 - IsRainDay = 1 if PRCP_in > 0; = 0 otherwise
 - Month = 1–12 (integer)

Why IsRainDay? A rain day is an event in the Poisson sense — it either happens or it doesn’t on a given day. Counting rain days per month gives us k (the observed event count). Averaging those counts across months gives us λ (the estimated rate).

Poisson validity check (ungraded): Before applying POISSON.DIST, verify the rain-day data meets the three Poisson conditions: (1) discrete events — each day is either a rain day or not; (2) independence — whether it rained yesterday does not change today’s probability; (3) constant rate — we assume λ is stable across the 2013–2022 period. Condition 3 is the one most worth scrutinizing: your seasonal analysis in Sections A and B will reveal whether a single λ is adequate, or whether seasonal rates are needed instead.

Section A — Marginal Event Rate (~25 min)

All Section A answers use COUNTIF, COUNTIFS, or POISSON.DIST. Round percentages to 3 decimal places.

A1 — Total Rain Days

How many days in 2013–2022 recorded any precipitation (IsRainDay = 1)?

A2 — Overall Lambda

The dataset covers exactly 120 months (10 years $\times$ 12 months). Compute the overall average rain days per month:

$$\lambda_{\text{overall}} = \text{A1 value} / 120$$

λ is the single most important number in a Poisson model. It is both the mean and the variance of the distribution. A system designed for the average (λ) will fail 50% of the time — because half of all months will exceed the mean.

A3 — P(Dry Month): PMF at $k = 0$

What fraction of months have zero rain days, if rain follows a Poisson distribution with your λ_{overall} ?

Multiply by 100 to express as a percentage.

`POISSON.DIST(k, lambda, FALSE)` computes the probability mass function — the probability of exactly k events. `FALSE` selects the PMF; `TRUE` selects the CDF.

A4 — P(Exactly 5 Rain Days): PMF at $k = 5$

A5 — P(At Most 5 Rain Days): CDF at $k = 5$

`TRUE` gives the cumulative distribution function — $P(X \leq k)$. Notice how $A5 > A4$: the CDF accumulates all probabilities from $k = 0$ up through $k = 5$.

A6 — Summer Rain Days

How many of the 3,652 days are in Summer (Season = “Summer”) AND are rain days (IsRainDay = 1)?

A7 — Summer Lambda

The dataset has 30 Summer months (10 years $\times$ 3 summer months per year). Compute:

$$\lambda_{\text{summer}} = \text{A6 value} / 30$$

A8 — P(Exactly 10 Rain Days in a Summer Month)

Compare the Summer λ you computed in A7 to the overall λ from A2. The ratio shows how much more rain-active summer is. This is the same conditioning insight from Week 3 — now formalized as a rate.

Section B — Seasonal Poisson Analysis

Report percentages to 3 decimal places, λ values to 2 decimal places.

B1 — Spring Rain Days

How many Spring days (Season = “Spring”) are rain days?

B2 ★ — Spring Lambda

The dataset has 30 Spring months.

Col D: number of Spring months in the dataset · Col E: $\lambda_{\text{spring}} = B1 / 30$

B3 — P(Dry Spring Month)

Compare to A3 (dry month at λ_{overall}). How does the probability of a completely dry month change in spring?

B4 — P(Unusually Wet Spring): $P(X \geq 5)$

An “unusually wet” spring month is one with 5 or more rain days. At the spring rate:

The complement rule: $P(X \geq 5) = 1 - P(X \leq 4)$. This is how you compute “exceedance probabilities” from a Poisson CDF.

B5 — July Rain Days

How many days in July (Month = 7) are rain days?

B6 ★ — July Lambda

July λ is computed per year (10 Julys in the dataset), not per month.

Col D: number of complete Julys in the dataset · Col E: $\lambda_{\text{july}} = B5 / 10$

July λ should be dramatically higher than spring λ . This is Tucson’s North American Monsoon. Warm moist air from the Gulf of California drives intense convective storms in July and August — a phenomenon so reliable it has its own name and its own engineering design requirements.

Engineering Context: ASCE 7-22 specifies design loads using return periods — the average time between exceedance events. Under the Poisson model:

$$P(\geq 1 \text{ event in } N \text{ years}) = 1 - e^{-(N/T)} = 1 - e^{-(10/25)} = 32.97\%$$

B7 — P(Dry July): The Monsoon Test

If your answer is less than 0.01%, that is correct. A dry July in Tucson is nearly impossible — the monsoon produces rain on 10+ days in a typical July. An engineer who designs outdoor drainage assuming “it might not rain” in July is designing for the wrong distribution.

B8 — P(Extreme July): $P(X > 15)$

An extreme July has more than 15 rain days. At λ_{july} :

ArcE Framing: ASHRAE 90.1-2022 requires roof drainage systems sized for design storm intensity — not the annual average. For Tucson, the 100-year, 1-hour intensity is approximately 2.5 in/hr (NOAA Atlas 14).

B9 — Winter Rain Days

How many Winter days (Season = “Winter”) are rain days?

B10 ★ — Winter Lambda

Col D: number of Winter months in the dataset · Col E: $\lambda_{\text{winter}} = B9 / 30$

B11 — P(Dry Winter Month)

Compare B11 (winter) to B3 (spring) and A3 (overall). Which season has the highest probability of a completely dry month? Explain physically why.

B12 — P(Wet Winter Month): $P(X \geq 8)$

B13 — Return Period Calculation

A stormwater system is designed for the 25-year storm (return period $T = 25$ years). What is the probability that this design storm occurs at least once during a 10-year project analysis period?

Using the Poisson return-period formula:

$$P(\text{at least 1 in } N \text{ years}) = 1 - e^{(-N/T)}$$

Important: Use EXP() for this formula — not POISSON.DIST(). The approach column requires EXP.

Why EXP and not POISSON.DIST? This is $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{(-\lambda)}$, where $\lambda = N/T = 10/25 = 0.4$ events expected in 10 years. Writing it with EXP makes the exponential structure explicit and matches how engineers

Target shape for Chart 1

Figure 1: Target shape for Chart 1

write return-period calculations in practice. The result should surprise you — a 25-year storm is far from rare over a 10-year design life.

B14 ★ — Is an Above-Average July Common?

The July mean is $\lambda_{\text{july}} = 10.3$ rain days. You might expect that exactly half of all Julys are above average. Is that true?

Compute $P(X > 10 \mid \lambda_{\text{july}})$ — the probability that a randomly chosen July has more rain days than the mean:

Col D: $P(X \leq 10 \mid \lambda_{\text{july}})$ · Col E: $P(X > 10 \mid \lambda_{\text{july}})$

Since λ_{july} is not a whole number (10.3), the Poisson distribution is very slightly right-skewed — $P(\text{above mean})$ is slightly less than 50%. This asymmetry grows larger at small λ and shrinks toward 50% as $\lambda \rightarrow \infty$.

Required Charts

Insert both charts in your working Excel file.

Chart 1 — PMF Comparison: Spring vs. Summer (7 pts)

Build a PMF table for $k = 0-15$ with two columns: $P(X=k \mid \lambda_{\text{spring}})$ and $P(X=k \mid \lambda_{\text{summer}})$, using `=POISSON.DIST(k, lambda, FALSE)` with absolute references for lambda. Plot as a clustered column chart.

- Title: Poisson PMF — Spring vs. Summer, Tucson 2013–2022
- Axes: Rain Days per Month (k) on x, Probability on y (formatted as %)
- Two distinct colors with a legend

Chart 2 — Lambda by Season Profile (8 pts)

Build a 4-row table (Winter / Spring / Summer / Fall) with your seasonal λ values. For Fall, use `COUNTIFS(Season = "Fall", IsRainDay = 1) / 30`. Plot as a clustered column in meteorological order.

- Title: Mean Rain Days per Month by Season — Tucson 2013–2022
 - Y-axis: λ (rain days / month)
 - Data labels on each bar showing the λ value
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Target shape for Chart 2

Figure 2: Target shape for Chart 2

Memo Deliverable

Type your memo directly in the Written sheet of your Answer Sheet Excel file. The sheet has labeled yellow cells for each paragraph (P1, P2, P3). Press Alt+Enter within a cell to add new lines. No separate PDF file is needed.

To: CE 310 Instructor

From: [Your Name], [Your Major]

Date: [Date]

Subject: Week 4 Lab — Poisson Analysis of Tucson Precipitation, 2013–2022

Paragraph 1 — The Event Rate: State your overall λ , summer λ , and spring λ . Using the PMF results from Section A and B, describe what the Poisson model says about the likelihood of a “dry month” (zero rain days) in each season. Cite at least two specific PMF values.

Paragraph 2 — Exceedance and Design: Using your B8, B12, and B13 results, explain the concept of exceedance probability and its relationship to return periods. In your own words, explain why $P(X > 15 \text{ in July})$ and $P(25\text{-year storm in 10 years})$ are both Poisson questions, and what they have in common methodologically.

Paragraph 3 — Monsoon drainage design and building envelope:

A building envelope consultant is specifying roof drainage and window flashing for a medium office building in Tucson. Using your seasonal λ values, explain why the monsoon season (Summer/July) dominates the drainage design requirement while winter rainfall drives a different set of concerns (slow, sustained events vs. intense convective bursts). Reference the ASHRAE 90.1 design-storm context from the ArcE Framing callout. Cite at least one Poisson probability from Section B.

Submission: Save as CE310_W4_<NetID>.xlsx and upload to the Week 4 D2L dropbox. A 5-point deduction applies if your file cannot be matched to your student record.

Before You Submit

- ☐ Full name and NetID entered on the Answer Sheet
- ☐ File saved as .xlsx and named CE310_W4_<NetID>.xlsx
- ☐ Sheet tabs not renamed, reordered, or deleted
- ☐ Every answer typed into its designated cell in Col E — answers placed anywhere else receive no credit

- ☐ Your formula typed as text in Col C, with an apostrophe before the = — screenshots are not accepted
 - ☐ ★ questions also have their intermediate value in Col D
 - ☐ Uploaded to D2L before the deadline
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Grading Summary

Component	Points
Section A — A1–A8	16
Section B — B1–B14	60
Chart 1 — PMF comparison, Spring vs. Summer	7
Chart 2 — Seasonal lambda profile	8
Memo (3 paragraphs)	9
Total	100